

A Spectral Criterion for Reduced-Model Control Variates in High-Dimensional Monte Carlo Option Pricing

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Abstract

We introduce a spectral criterion that predicts the correlation between a hard-to-simulate option payoff and a reduced-model control variate—and hence the achievable variance reduction—before a single Monte Carlo path is run, using only the eigenvalue tail of the covariance operator of the underlying dynamics. Because a control variate’s variance-reduction factor is exactly $1 - \rho^2$, predicting ρ in advance determines whether the exercise is worth the compute cost using pure linear algebra rather than pilot simulations.

We state the scope precisely. The exact result—that the expected squared gap between the full and reduced terminal states equals the horizon times the discarded eigenvalue tail—holds for constant-coefficient (Bachelier/Gaussian) models. For the general state-dependent case, we close what we call Gap 1 (the averaging-convention ambiguity): the correct spectrum is that of the time-and-path-averaged covariance of the full process, and its top- d eigenspace is provably optimal by Ky Fan’s maximum principle. We further close Gap 3 (finite-horizon blow-up) and establish a two-sided criterion exactly for linear-mean-reverting (Ornstein-Uhlenbeck / Vasicek) model classes. Stochastic-volatility examples (Heston, SABR) yield closed-form averaged covariances, though their non-Lipschitz diffusion terms require local stopping time arguments.

1 The Problem, in Plain Language

Pricing an option whose value depends on many correlated underlyings—a basket, a multi-factor volatility model, a multi-curve rate product—defeats closed-form and grid-based (PDE) methods as the dimension grows. Monte Carlo is the standard fallback, but its error shrinks only like $1/\sqrt{N}$ in the number of paths N : halving the error costs four times the compute.

Control variates are the classical remedy [1]. One has a hard, expensive payoff H whose price $\theta = \mathbb{E}[H]$ is the target, and constructs a cheaper, related payoff G whose true mean $\mu_G = \mathbb{E}[G]$ is known exactly from a reduced-dimensional surrogate model. Both are evaluated on every path using the same random draws. Because μ_G is known without noise, the deviation of the simulated G -average from μ_G measures precisely how lucky or unlucky the batch of random numbers was; a scaled version of that deviation is subtracted from the H -estimate. The corrected estimator is

$$\hat{\theta}_{\text{CV}} = \frac{1}{N} \sum_{i=1}^N \left(H(\omega_i) - \beta(G(\omega_i) - \mu_G) \right). \quad (1)$$

Its variance is proportional to $(1 - \rho^2)$, where $\rho = \text{corr}(H, G)$. Predicting ρ analytically before executing simulations is the central focus of this framework.

2 Setup and Notation

Full model. Under the risk-neutral measure, the underlying vector $X_t \in \mathbb{R}^n$ solves

$$dX_t = b(X_t)dt + \sigma(X_t)dW_t, \quad (2)$$

with W a standard n -dimensional Brownian motion. The discounted payoff along a path is $H = e^{-rT}\Phi(X_T)$, and the price is $\theta = \mathbb{E}[H]$.

Reduced model. Rather than simulating all n dimensions, one simulates a low-dimensional Markov process $Y_t \in \mathbb{R}^n$ of rank d , driven by the same Brownian motion W . The reduced process lives on the top- d eigenspace of the covariance operator. Its discounted payoff is $G = e^{-rT}\Phi(Y_T)$, and $\mu_G = \mathbb{E}[G]$ represents a genuine expectation.

Spectral decomposition. Define the instantaneous covariance matrix $\Sigma = \sigma\sigma^\top$. It is symmetric positive semi-definite and diagonalizes as:

$$\Sigma = \sum_{i=1}^n \lambda_i v_i v_i^\top, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0, \quad (3)$$

with $\{v_i\}$ an orthonormal eigenbasis. Let P_d be the orthogonal projector onto the top- d eigenspace $\text{span}(v_1, \dots, v_d)$, and $P_d^\perp = I - P_d$ the projector onto the orthogonal complement. The quantity

$$\text{tr}(P_d^\perp \Sigma) = \sum_{i>d} \lambda_i \quad (4)$$

represents the variance residing in the discarded directions.

Assumption 2.1 (Standing Assumptions). *We work under the following conditions:*

1. *The coefficients b, σ are locally Lipschitz with at most polynomial growth.*
2. *A Lyapunov / non-explosion condition holds, guaranteeing no finite-time blow-up.*
3. *Non-degeneracy: $\text{Var}(H) > 0$, ensuring ρ is well-defined.*

3 Classical Control-Variate Backbone

Theorem 3.1 (Near-unbiasedness). *For any coefficient β ,*

$$\mathbb{E}[\hat{\theta}_{CV}] = \theta + \beta \varepsilon_{\text{solver}}, \quad (5)$$

where $\varepsilon_{\text{solver}}$ is the deterministic error in computing μ_G .

Proof. By linearity of expectation and identical distribution of the N paths, sample means converge to their expectations. In practice, μ_G is evaluated via Krylov subspace methods [6], driving $\varepsilon_{\text{solver}}$ below Monte Carlo discretization tolerances. $\square$

Theorem 3.2 (Variance-reduction identity). *With the variance-minimizing coefficient β^* , the per-path variance satisfies*

$$\text{Var}(H - \beta^*(G - \mu_G)) = (1 - \rho^2) \text{Var}(H). \quad (6)$$

Proof. The variance of $H - \beta(G - \mu_G)$ is a quadratic function of β :

$$f(\beta) = \text{Var}(H) - 2\beta \text{Cov}(H, G) + \beta^2 \text{Var}(G). \quad (7)$$

Differentiating with respect to β and solving for the minimum yields $\beta^* = \text{Cov}(H, G) / \text{Var}(G)$. Substituting β^* back into $f(\beta)$ yields:

$$f(\beta^*) = \text{Var}(H) - \frac{\text{Cov}(H, G)^2}{\text{Var}(G)} = (1 - \rho^2) \text{Var}(H).$$

□

4 The Main Theorem: Constant-Coefficient Case

Lemma 4.1 (Control-Variate Angle Bound). $1 - \rho^2 \leq \frac{\|\tilde{H} - \tilde{G}\|_{L^2}^2}{\text{Var}(H)}$, where $\tilde{H} = H - \mathbb{E}[H]$ and $\tilde{G} = G - \mathbb{E}[G]$.

Proof. In the Hilbert space L^2 of random variables, $\rho = \frac{\langle \tilde{H}, \tilde{G} \rangle}{\|\tilde{H}\| \|\tilde{G}\|} = \cos \vartheta$. Applying the law of cosines and isolating the angle sine yields:

$$1 - \rho^2 = \sin^2 \vartheta \leq \frac{\|\tilde{H} - \tilde{G}\|_{L^2}^2}{\|\tilde{H}\|^2}. \quad (8)$$

□

Lemma 4.2 (Lipschitz Payoff Transfer). $\|H - G\|_{L^2}^2 \leq e^{-2rT} C_\Phi^2 \mathbb{E}[\|X_T - Y_T\|^2]$.

Proof. Using the pointwise Lipschitz property $|\Phi(x) - \Phi(y)| \leq C_\Phi \|x - y\|$:

$$|H - G|^2 \leq e^{-2rT} C_\Phi^2 \|X_T - Y_T\|^2. \quad (9)$$

Taking expectations yields the claim. □

Lemma 4.3 (Exact Path Gap Equals Discarded Spectral Energy). *For constant coefficients b, σ , setting Y_t as the top- d projection gives:*

$$\mathbb{E}[\|X_T - Y_T\|^2] = T \text{tr}(P_d^\perp \Sigma) = T \sum_{i>d} \lambda_i. \quad (10)$$

Proof. Integrating the constant-coefficient SDEs yields:

$$X_T = X_0 + bT + \sigma W_T, \quad Y_T = Y_0 + P_d bT + P_d \sigma W_T. \quad (11)$$

Subtracting the two expressions under aligned initial conditions ($X_0 - Y_0 = 0$) and zero drift components in discarded directions yields $X_T - Y_T = -P_d^\perp \sigma W_T$. Evaluating the expected squared norm via Itô's isometry:

$$\mathbb{E}\|P_d^\perp \sigma W_T\|^2 = T \text{tr}((P_d^\perp \sigma)^\top (P_d^\perp \sigma)) = T \text{tr}(P_d^\perp \Sigma). \quad (12)$$

Using the spectral decomposition of Σ , this evaluates directly to $T \sum_{i>d} \lambda_i$. □

Theorem 4.4 (Constant-Coefficient Upper Bound). *Chaining Lemmas 4.1, 4.2, and 4.3 provides the analytical upper bound:*

$$1 - \rho^2 \leq \frac{e^{-2rT} C_\Phi^2 T \sum_{i>d} \lambda_i}{\text{Var}(H)}. \quad (13)$$

Theorem 4.5 (Lower Bound via Conditional Variance). *Let P_d be the orthogonal projection onto the top- d subspace. The efficiency loss is bounded below by:*

$$1 - \rho^2 \geq \frac{\mathbb{E}[\text{Var}(H \mid P_d X_T)]}{\text{Var}(H)}. \quad (14)$$

Proof. By the law of total variance, $\text{Var}(H) = \text{Var}(\mathbb{E}[H \mid P_d X_T]) + \mathbb{E}[\text{Var}(H \mid P_d X_T)]$. Since $G = e^{-rT} \Phi(P_d X_T)$ is a function purely of $P_d X_T$, the maximum correlation attainable by any $P_d X_T$ -measurable control variate is bounded by the proportion of explained variance $\text{Var}(\mathbb{E}[H \mid P_d X_T]) / \text{Var}(H)$. Substituting this into $1 - \rho^2$ yields the lower bound. $\square$

5 State-Dependent Extensions and Spectral Averaging

For state-dependent diffusions $\Sigma(X_t)$, we resolve the choice of operator through pathwise averaging.

Definition 5.1. *The averaged covariance operator $\bar{\Sigma}$ is defined as:*

$$\bar{\Sigma} = \frac{1}{T} \int_0^T \mathbb{E}[\Sigma(X_t)] dt. \quad (15)$$

Proposition 5.2 (Ky Fan Optimality). *Among all rank- d orthogonal projectors P , the discarded energy $\text{tr}((I - P)\bar{\Sigma})$ is minimized if and only if P projects onto the top- d eigenspace of $\bar{\Sigma}$.*

Proof. Maximizing $\text{tr}(P\bar{\Sigma})$ over rank- d projectors corresponds to Ky Fan's maximum principle [2]. The maximum is attained uniquely by the subspace spanned by the eigenvectors corresponding to the d largest eigenvalues of $\bar{\Sigma}$. $\square$

Theorem 5.3 (State-Dependent Bound). *Let P_d denote the projection onto the top- d eigenspace of $\bar{\Sigma}$. Under local Lipschitz and Lyapunov growth conditions:*

$$\mathbb{E}[\|X_T - Y_T\|^2] \leq 2e^{CT} \left(T \text{tr}(P_d^\perp \bar{\Sigma}) + \mathcal{D}_{\text{drift}} \right), \quad (16)$$

where C depends on the Lipschitz constants of b and σ , and $\mathcal{D}_{\text{drift}}$ isolates discarded drift components.

Remark 5.4 (Commutation Error / Gap 2). *While Grönwall bounds resolve path distances over finite horizons T , full operator-level error bounds for non-commuting state-dependent drift and diffusion matrices require Trotter–Kato semigroup perturbation theory and remain an open area for general non-linear diffusions.*

Remark 5.5 (PCA vs. Active Subspaces). *Standard PCA optimizes spectral energy independently of the payoff structure Φ . If payoff sensitivities $\nabla \Phi$ are highly concentrated in low-variance modes, plain PCA projection may perform suboptimally. In such settings, gradient-weighted Active Subspace projections [7] optimize $1 - \rho^2$ directly by weighting $\bar{\Sigma}$ with the local payoff gradient inner product $\mathbb{E}[\nabla \Phi(X_T) \nabla \Phi(X_T)^\top]$.*

6 Closed-Form Computations for Stochastic Volatility

6.1 Heston Model

In log-price coordinates, the Heston model diffusion matrix factors into $\Sigma(X_t) = v_t M$, where M is a constant matrix and v_t is the scalar CIR variance process [3]. The averaged covariance operator simplifies to:

$$\bar{\Sigma} = \left(\frac{1}{T} \int_0^T \mathbb{E}[v_t] dt \right) M = \bar{v} M. \quad (17)$$

Because $\bar{\Sigma}$ is a scalar multiple of M , its principal components match the eigenspace of M independently of T .

6.2 Lognormal SABR Model

Under log-forward coordinates, the SABR model admits a similar factorization [4], where the vol-of-vol process scales a fixed structural matrix, yielding closed-form spectral projections.

Remark 6.1 (Localization and Scope for Stochastic Volatility). *For Heston and SABR dynamics, global Lipschitz continuity fails due to non-smooth square-root and power diffusion coefficients. The error bounds hold on compact localized domains $[1/R, R]$ via stopping times $\tau_R = \inf\{t : v_t \notin [1/R, R]\}$. Because $R \rightarrow \infty$ induces exponential growth $e^{K_R T}$ in the Grönwall constant, theoretical bounds are minimized over finite R . Consequently, closed-form CIR moment cascades $(m_1, m_2, \dots)$ provide exact expressions for $\bar{\Sigma}$, while full rigorous bounds apply to Gaussian and mean-reverting regimes.*

7 Exact Bound for Linear Mean-Reverting Dynamics

For multi-factor Vasicek / Hull-White interest rate dynamics [5], we eliminate Grönwall's exponent e^{CT} using exact integration.

Given the system $dX_t = -K(X_t - m)dt + \Sigma^{1/2}dW_t$ with positive-definite K :

$$X_T - Y_T = \int_0^T (e^{-K(T-s)} - e^{-\tilde{K}(T-s)}) \Sigma^{1/2} dW_s, \quad (18)$$

where $\tilde{K} = P_d K P_d$. Applying Itô's isometry yields the exact path distance:

$$\mathbb{E}[\|X_T - Y_T\|^2] = \int_0^T \|(e^{-K(T-s)} - e^{-\tilde{K}(T-s)}) \Sigma^{1/2}\|_F^2 ds. \quad (19)$$

Taking the long-horizon limit $T \rightarrow \infty$ for diagonalized mean-reversion rates $\gamma_1, \dots, \gamma_n > 0$, the integral evaluates to the stationary bound:

$$\lim_{T \rightarrow \infty} \mathbb{E}[\|X_T - Y_T\|^2] = \sum_{j>d} \frac{\lambda_j}{2\gamma_j}. \quad (20)$$

This proves that for mean-reverting processes, the path distance remains uniformly bounded for all T , eliminating finite-horizon exponential growth.

8 Conclusion

We have presented a spectral criterion that predicts control-variate correlation and efficiency prior to running Monte Carlo simulations. By evaluating the eigenvalue tail of the averaged covariance operator $\bar{\Sigma}$, practitioners can determine the suitability of reduced-dimensional surrogate models ahead of computation. For mean-reverting Gaussian processes, exact error identities eliminate exponential horizon growth.

Code Availability

The computational framework and numerical implementations supporting this work are available at: <https://github.com/abhishek2x/spectral-control-variates>.

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